

A Surrogate Model for Reducing the Computational Cost of Neuron Simulations

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Introduction

- Multi-compartment models represent the spatial morphology of a neuron as connected compartments, reproducing neuronal activity in fine detail.
- Each compartment is described by differential equations carrying **gate variables** (ion-channel open probabilities), as in the Hodgkin–Huxley (HH) model.

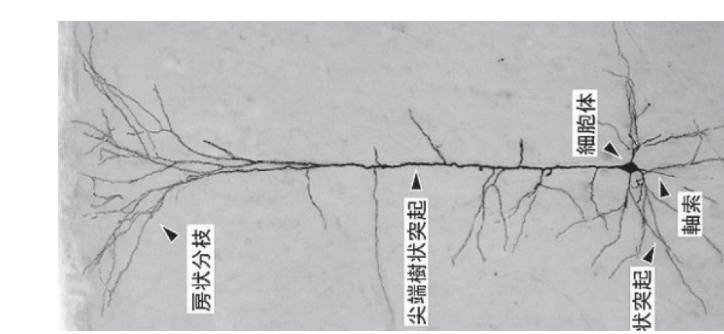
$$C_m \frac{dV}{dt} = -\bar{g}_L(V - E_L) - \bar{g}_{Na}m^3h(V - E_{Na}) - \bar{g}_Kn^4(V - E_K) + I_{\text{ext}}$$

HH membrane-potential equation (m, h, n : gate variables)

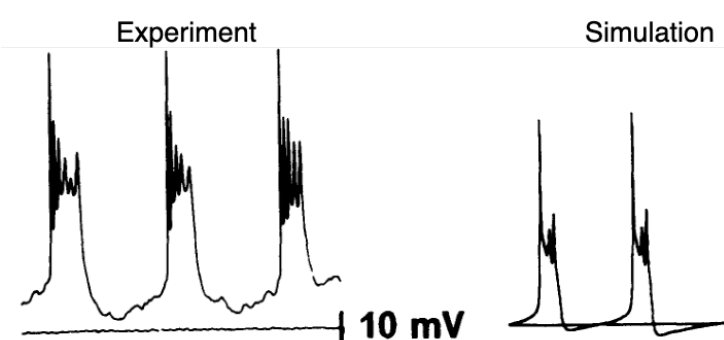
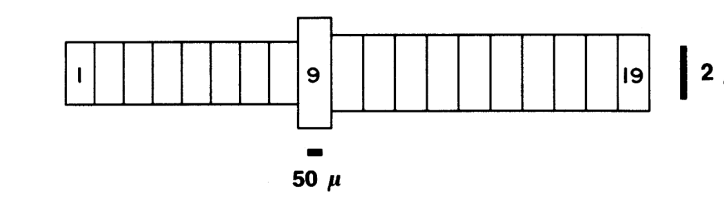
→ When many multi-compartment models are simulated in parallel (e.g. brain-scale simulation), the gate variables cause a **memory bottleneck**.

Goal

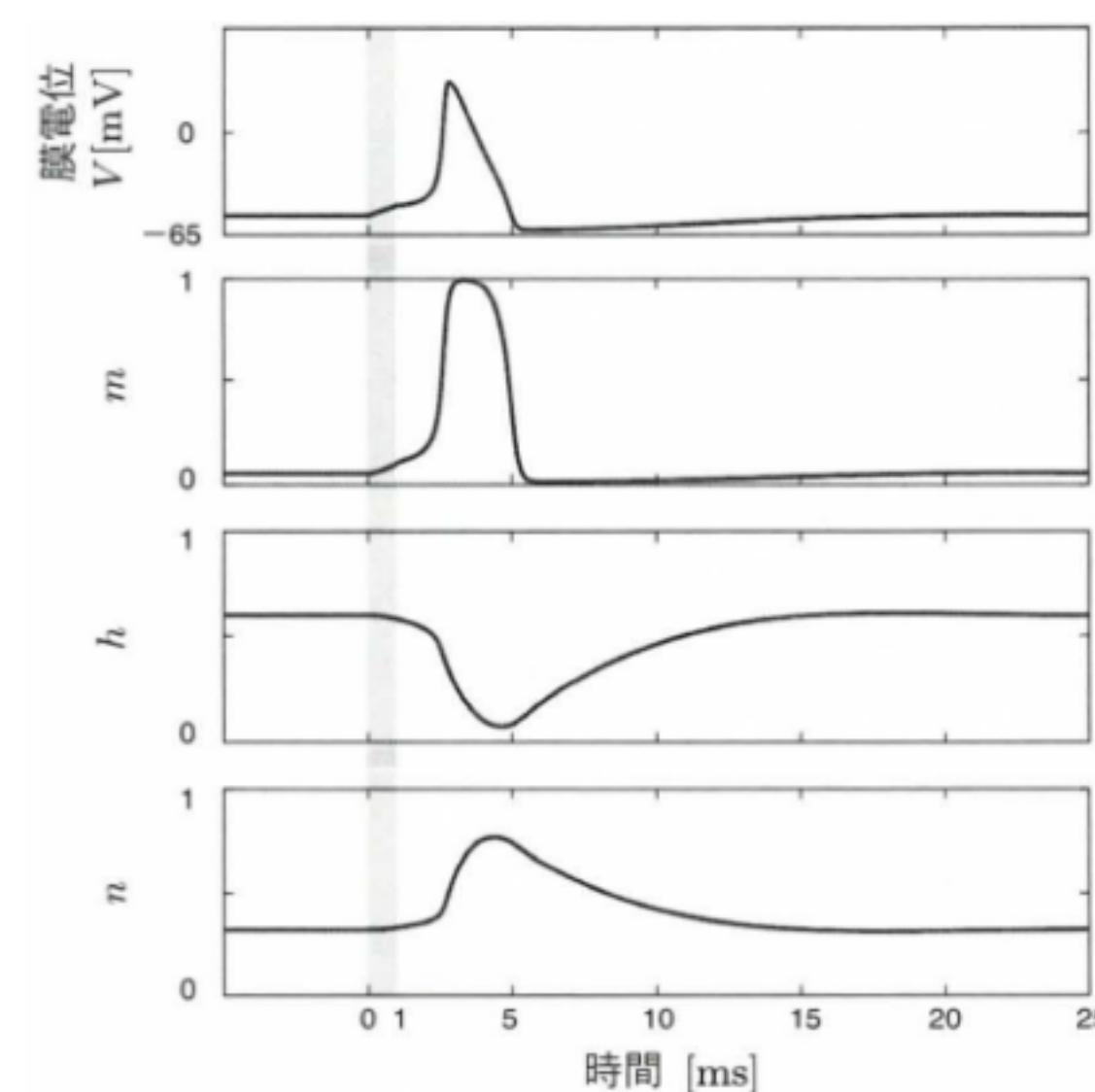
Build a surrogate model that reproduces the membrane-potential response of a multi-compartment neuron with **fewer state variables**.



Pyramidal neuron [1]



Spatial representation and simulation of a multi-compartment model [2]



HH membrane-potential response [3]

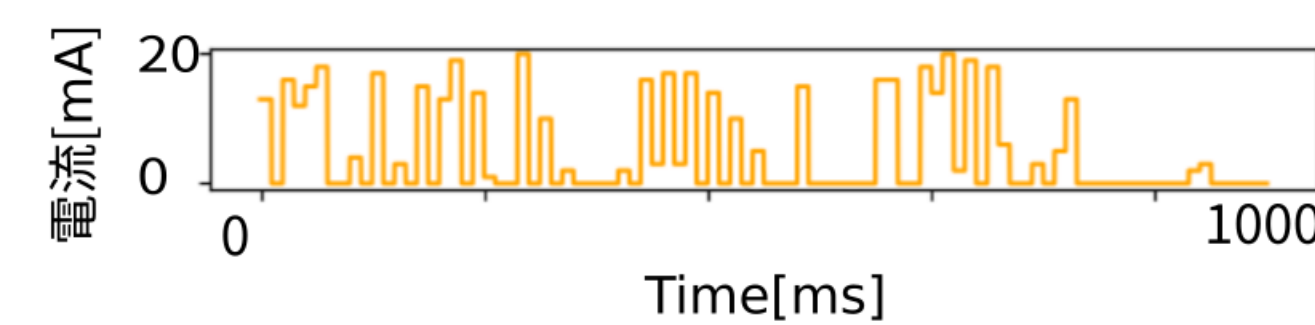
Methods

Two-axis surrogate framework

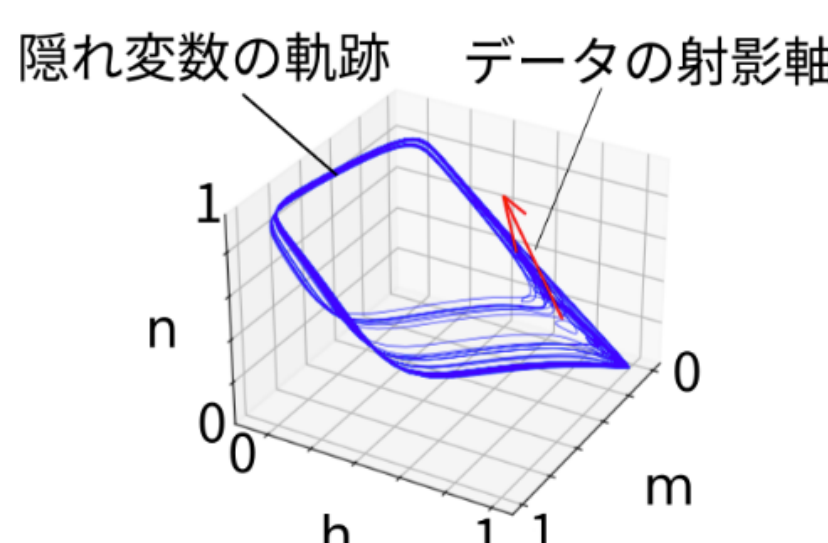
- Simulate the target model; collect time series of membrane potential V and gate variables.
- Compress** the gates into a low-dim latent $z \in \mathbb{R}^n$ with a **preprocessor** — PCA (linear) or an **autoencoder** (nonlinear).
- Identify** the latent dynamics with an **ansatz** — SINDy (full library) or **hybrid** (physics-separated).

$$\begin{cases} \frac{dV}{dt} = \underbrace{I_{\text{ion}}^{\text{phys}}(V, z)}_{\text{known physics}} + I_{\text{ext}} \\ \frac{dz}{dt} = \underbrace{\Theta(z, V)\xi}_{\text{SINDy-identified}} \\ z = \text{AE}_{\text{enc}}(\text{gates}), \quad \text{gates} = \text{AE}_{\text{dec}}(z) \end{cases}$$

The **hybrid** ansatz keeps the **known ionic physics** and lets SINDy identify only the **latent** dynamics — a direct answer to last year’s over-large (41-term) library that contained many non-physical terms.



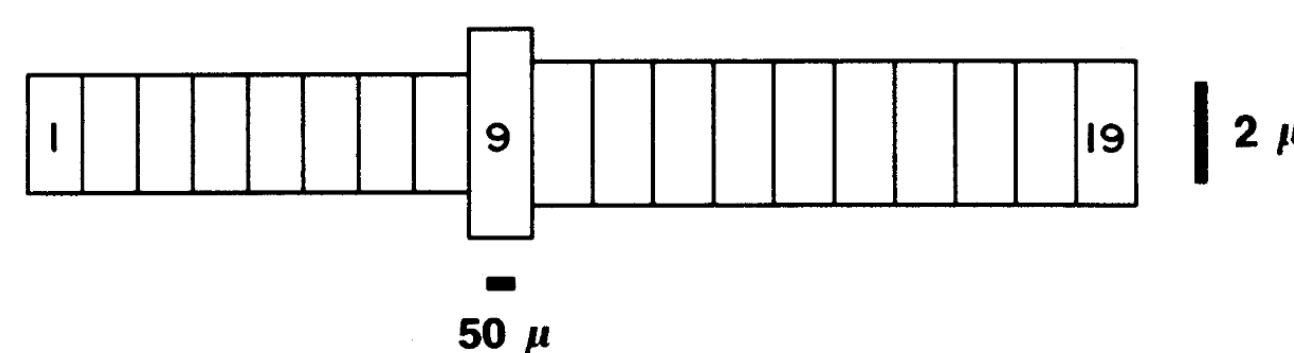
Random 10 ms pulse currents used to drive the simulation



Gate compression into latent space

Target: Traub CA3 pyramidal cell

(a 19-compartment model — a realistic multi-compartment neuron, replacing last year’s toy 3-compartment model)



We replace the **soma compartment** with the **surrogate**, so that the ionic gate variables of the soma are represented by the low-dimensional latent z while the compartment coupling stays physical.

- preprocessor $\in \{\text{PCA}, \text{AE}\}$
- ansatz $\in \{\text{SINDy}, \text{hybrid}\}$
- latent dim $n = 2$ (HH) / 3 (Traub)

SINDy

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} 1 & x & y & z & x^2 & xy & xz & y^2 & z^2 & \xi_1 & \xi_2 & \xi_3 \end{bmatrix} \begin{bmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \end{bmatrix}, \quad \Theta(X)$$

$\dot{X} = \Theta(X)\Xi$

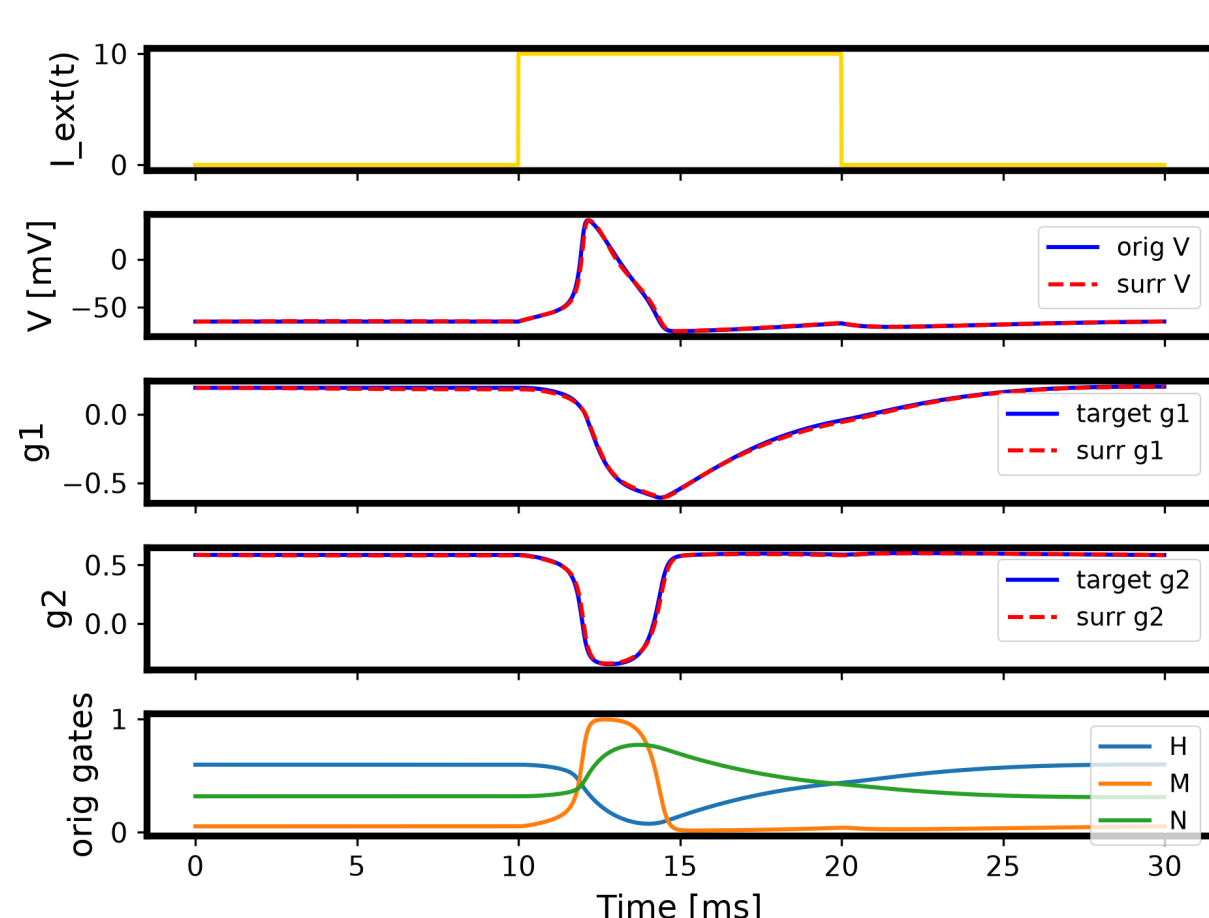
X : time-series data

$\Theta(X)$: library evaluated at X

Solve $\dot{X} = \Theta(X)\Xi$ as a sparse regression for the coefficient matrix Ξ [4]

Results and Discussion

① Proof of concept — single HH soma

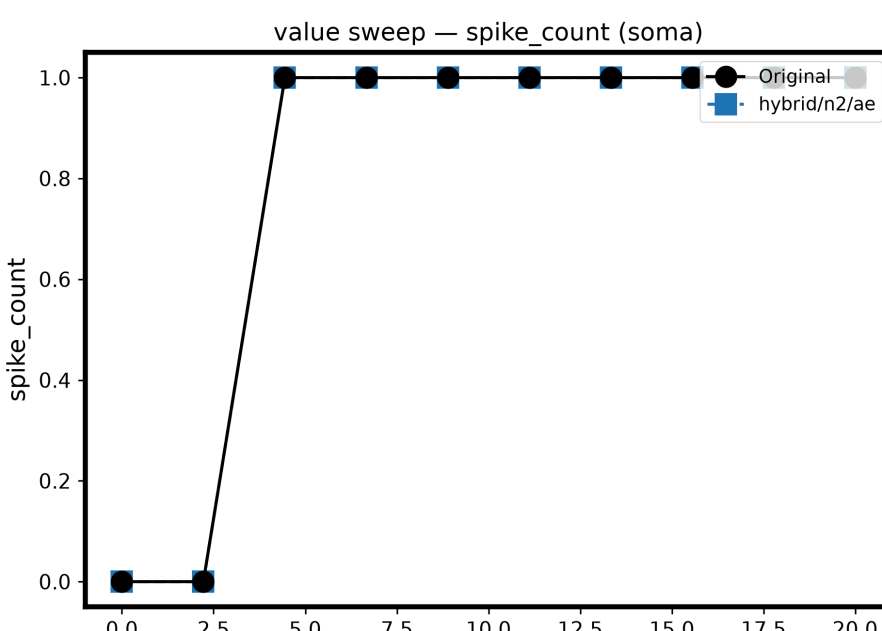


HH surrogate (AE, $n = 2$, hybrid): V , the two AE latents g_1, g_2 , and the original 3 gates. Original (blue) vs. surrogate (red, dashed) overlap almost exactly.

- 3 gates (m, h, n) → 2 latents**: the AE compresses the gates into a 2-D latent, yet the surrogate reproduces the spike with **near-zero error**.
- The hybrid ansatz retains the ionic physics, so the identified latent dynamics stay compact and stable.

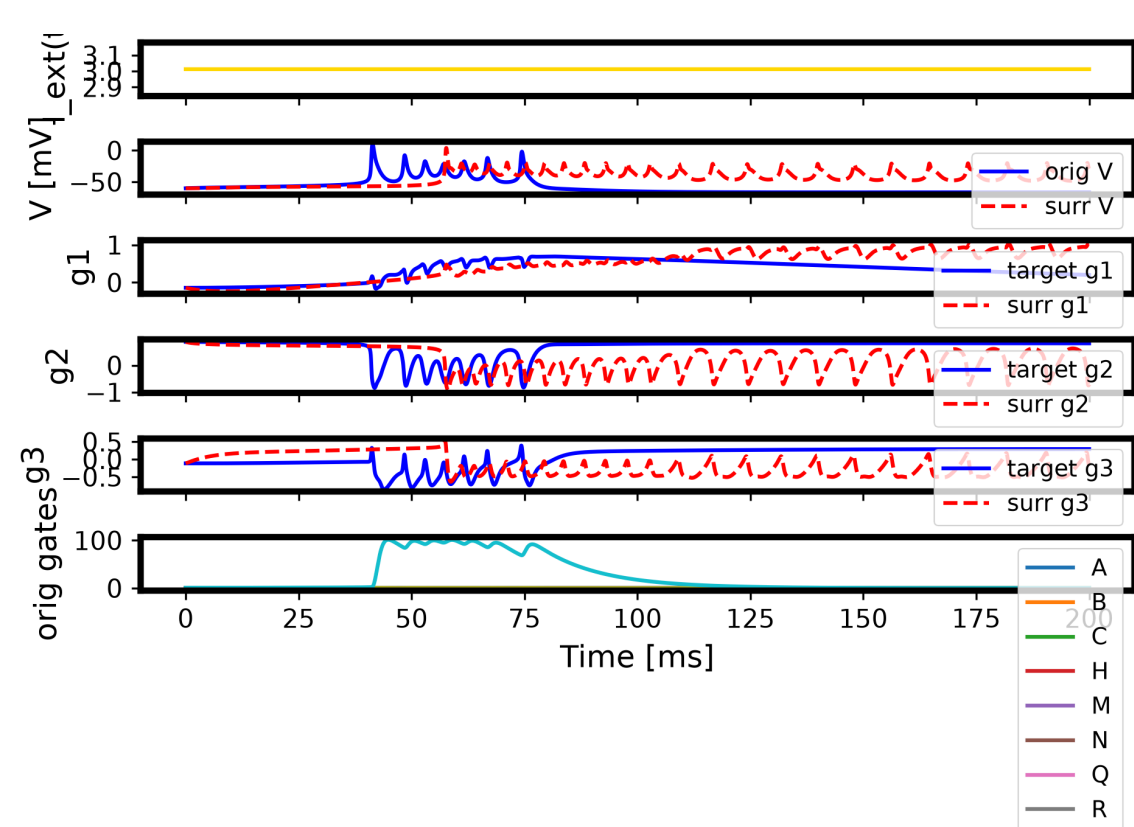
Waveform match

- spike-shape corr = **1.000**
- AP amplitude err = **0.5 mV**
- RMSE = **1.0 mV**



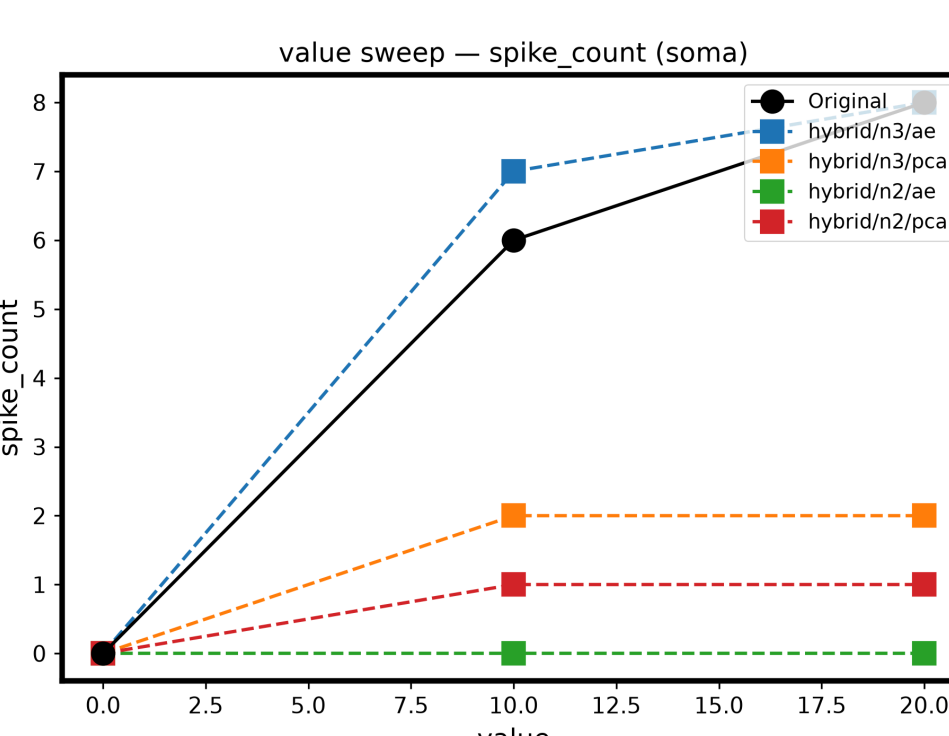
Spike count vs. input amplitude — matches the original over the whole range (not overfit to one stimulus).

② Scaling up — Traub 19-compartment (soma replaced)



Traub19 soma replaced (AE, $n = 3$, hybrid). V , the 3 latents, and the original soma gates. Original (blue) vs. surrogate (red, dashed).

- The **same AE + hybrid framework** transfers to a realistic multi-compartment model **without diverging**, firing in bursts and tracking the input-driven **spike count** (spike-shape corr = 1.000).
- It captures the **qualitative structure** — firing threshold and burst onset shift with input amplitude, as in the original.
- Remaining gap**: the post-burst **decay/termination** is not yet reproduced — the surrogate keeps oscillating.



Spike count across surrogate variants. Only **AE + $n = 3$** tracks the original; PCA and $n = 2$ collapse — motivating the nonlinear AE and the $n = 3$ latent.

Conclusion

- For a single HH neuron, the **AE + hybrid** surrogate reproduces the spike waveform with **near-zero error** while halving+ the state dimension ($4 \rightarrow 2$).
- The same framework **scales to a realistic 19-compartment Traub model** without diverging and captures the qualitative firing behaviour, though not yet quantitatively.

→ Future work

Reproduce the post-burst decay dynamics of the multi-compartment surrogate, and further refine the physics-informed library.

Code link

Code for this study is available at <https://github.com/MunechikaHaruki/SINDyNeuroSurrogate>

Bibliography

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